

2020 AMC 10A Problem 8 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 8

Problem:

What is the value of

$$1 + 2 + 3 - 4 + 5 + 6 + 7 - 8 + \cdots + 197 + 198 + 199 - 200?$$

Answer Choices:

- A. 9800
- B. 9,900
- C. 10,000
- D. 10,100
- E. 10,200

Solution:

Looking at the numbers, you see that every set of 4 has 3 positive numbers and 1 negative number. Calculating the sum of the first couple of sets gives us

$$2 + 10 + 18 \dots + 394$$

Clearly, this pattern is an arithmetic sequence. By using the formula we get

$$\frac{2 + 394}{2} \cdot 50 = \boxed{(B) 9900}.$$

OR

Note that the original expression is equal to

$$(1 + 2 + 3 + \cdots + 199 + 200) - 2(4(1 + 2 + 3 + \cdots + 49 + 50))$$

Since the sum of the first n positive integers is $\frac{n(n+1)}{2}$, this is equal to

$$\frac{200(201)}{2} - 2\left(4\left(\frac{50(51)}{2}\right)\right)$$

which can be simplified as

$$20100 - 4(50)(51) = 20100 - 10200 = \boxed{(B) 9900}.$$

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